Amrita Vishwa Vidyapeetham
Amrita School of Computing, Amritapuri
BTech Computer Science and Engineering (AI)

22AIE311 Software Engineering (Project based)

LAB SHEET 2

CO	Course Outcomes
CO01	Understand the basic principles of Software Engineering.
CO02	Understand how to choose the appropriate SDLC models depending on the user requirements.
CO03	Understand the concept of Requirements Engineering.
CO04	Understand the concept of Software Design.
CO05	Understand how to apply the knowledge, techniques, and skills in the development of software and its maintenance.

ELABORATION PHASE OF SOFTWARE DEVELOPMENT

Objective

To understand and perform the Elaboration Phase activities by modeling the system using Use Case Diagrams, Activity Diagrams, and Data Flow Diagrams (DFD Level 0, 1, and 2) and class diagram of ***your project.***

Group Number: 4

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PART A – USE CASE MODELING [CO04]

Task 1: Identify Actors

List all external entities interacting with the system.

Sl. No	Actor Name	Description
1	Citizen / Student	Uploads documents and checks the authenticity status
2	Issuer (University / Govt Authority)	Issues, registers, and revokes documents
3	Verifier (Employer / Organisation)	Verifies the authenticity of submitted documents
4	Admin	Manages system users, roles, and monitors fraud
5	Blockchain Network	Stores document hash and verification data
6	AI & Forensic System	Performs fraud detection and analysis

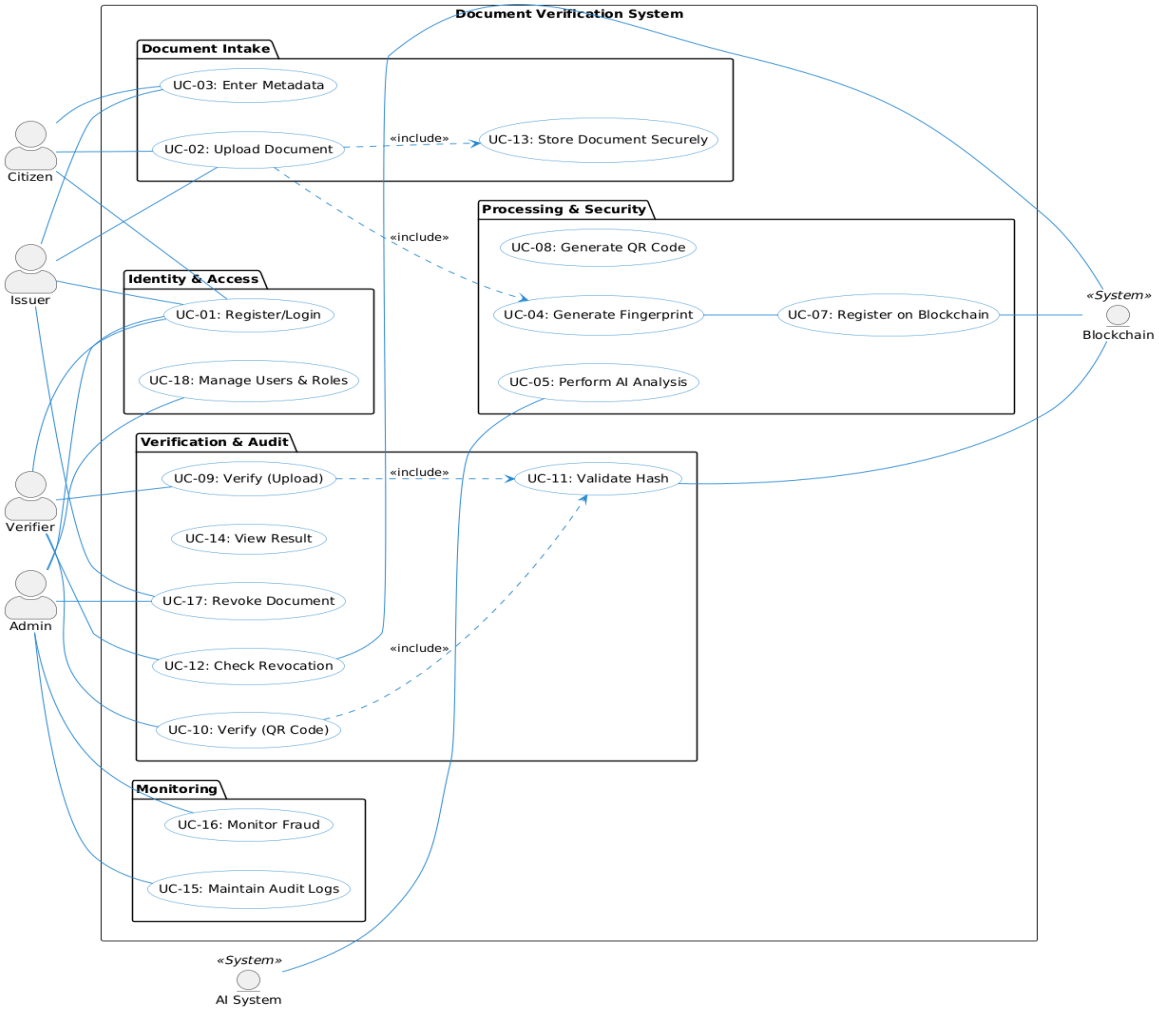
Task 2: Identify Use Cases

List the major functionalities of the system.

Use Case ID	Use Case Name	Actor(s)	Description
UC-01	Register/Login	All Users	Allows users to access the system securely
UC-02	Upload Document	Citizen / Issuer	User uploads document for verification
UC-03	Enter Metadata	Citizen / Issuer	User enters details like issuer, type, date
UC-04	Generate Fingerprint	System	Creates cryptographic hash of document
UC-05	Perform AI Analysis	AI System	Detects tampering and AI-generated content

UC-06	View Authenticity Score	Citizen / Verifier	Displays fraud risk and authenticity level
UC-07	Register Document on Blockchain	Issuer / System	Stores document hash and metadata
UC-08	Generate QR Code	System	Generates verification QR proof
UC-09	Verify Document (Upload)	Verifier	Verifies document via re-upload
UC-10	Verify via QR Code	Verifier	Scans QR to validate document
UC-11	Validate Hash with Blockchain	System	Compares uploaded doc with blockchain record
UC-12	Check Revocation Status	Verifier / System	Checks if document is revoked
UC-13	Store Document Securely	System	Saves encrypted document in storage
UC-14	View Verification Result	Citizen / Verifier	Displays final verification outcome
UC-15	Maintain Audit Logs	System / Admin	Logs all activities securely
UC-16	Monitor Fraud Patterns	Admin	Detects suspicious behavior
UC-17	Revoke Document	Issuer / Admin	Marks document as invalid
UC-18	Manage Users & Roles	Admin	Controls access and permissions

Task 3: Draw Use Case Diagram



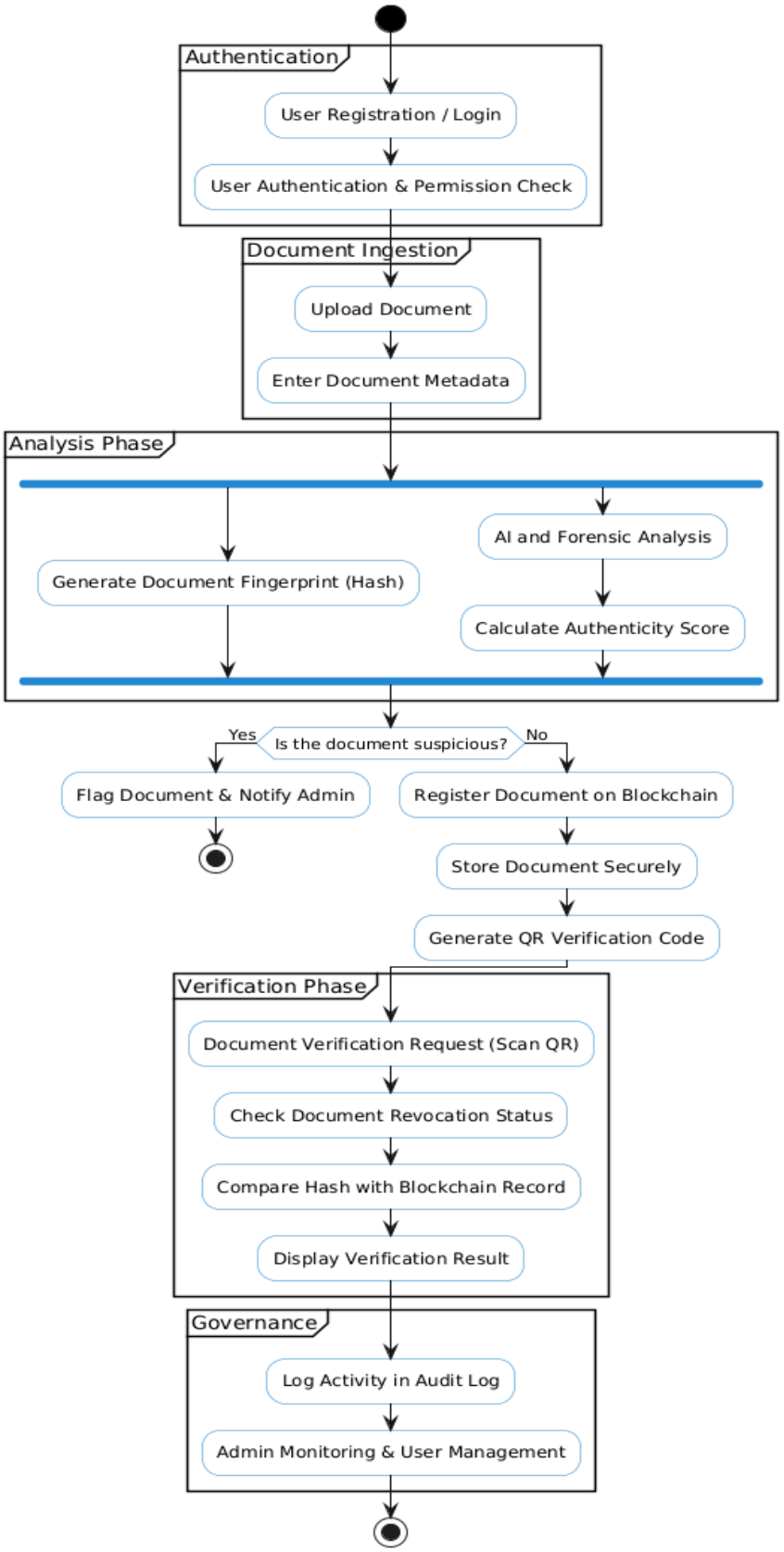
PART B – ACTIVITY MODELING [CO04]

Task 4: Identify Activities

1. User Registration / Login : Users, issuers, verifiers and administrators log into the system through the web portal.
2. User Authentication : the system validates the user and checks role permissions.
3. Upload Document : Authorized Issuer uploads document into the system.
4. Enter Document MetaData : Metadata such as issuer name, document type, issue date, and owner information are entered.
5. Generate Document Fingerprint(Hash) : The system generates a unique cryptographic hash of the uploaded document.
6. AI and Forensic Analysis : the system performs tamper detection, AI-generated content detection and metadata validation.
7. Calculate Authenticity Score : The AI module generates a risk score based on forensic analysis.

8. DECISION - Is the document suspicious? : If suspicious, flag the document and notify the admin. Else, continue processing.
9. Register Document on Blockchain : The document hash is recorded on the permissioned blockchain ledger.
10. Store document : The encrypted document is stored in secure storage.
11. Generate QR Verification Code : The system generates a QR code linked to the blockchain record.
12. Check Document Revocation Status : The system verifies whether the issuer revoked the document.
13. Document Verification Request : Verifier opens the document and scans the code.
14. Compare Hash with Blockchain Record : The system recomputes the document hash and compares it with the blockchain entry.
15. Display Verification Result : the system displays authenticity score, verification report and risk report.
16. Log Activity in Audit Log : Every action is recorded in the blockchain audit log.
17. Admin Monitoring : the admin monitors suspicious activities and manages system users.

Task 5: Draw Activity Diagram

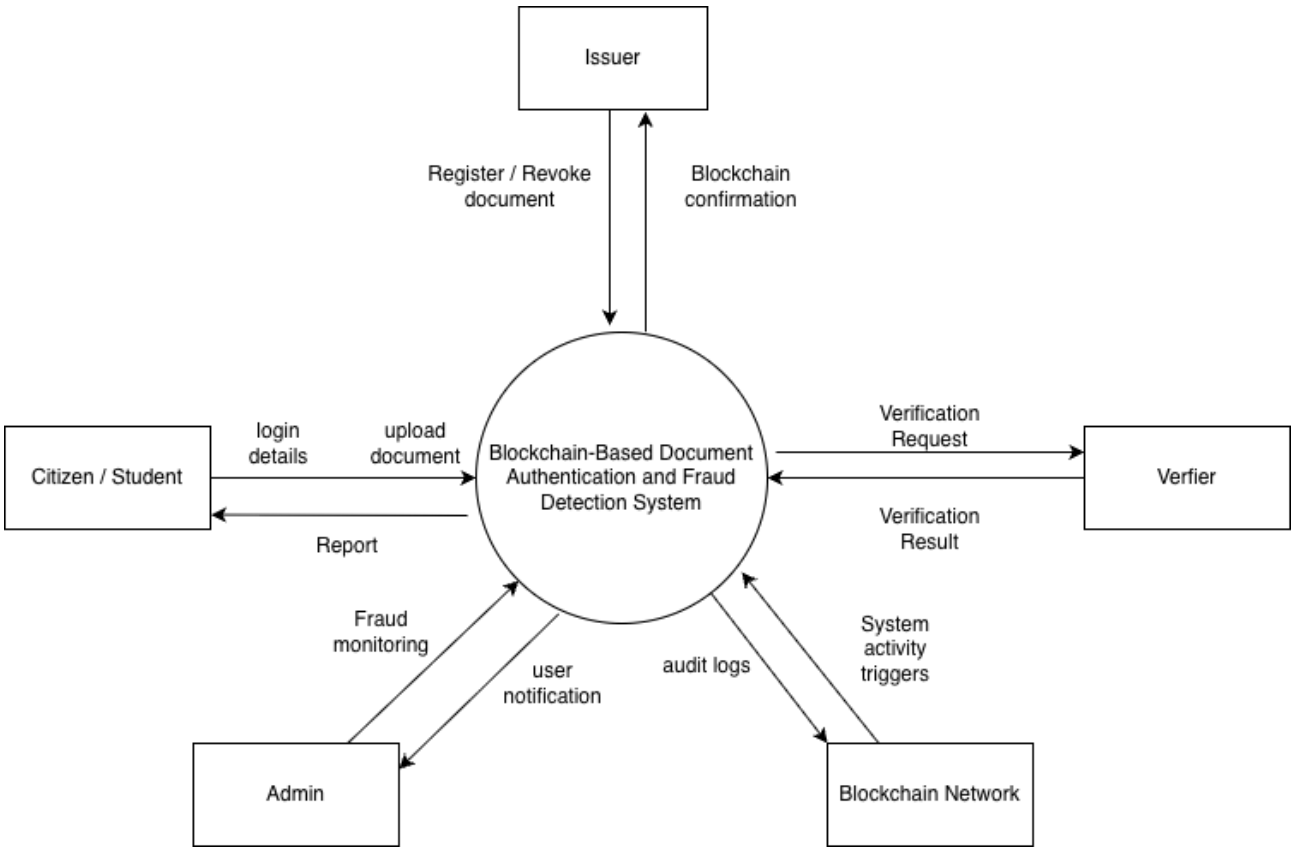


PART C – DATA FLOW MODELING [CO04]

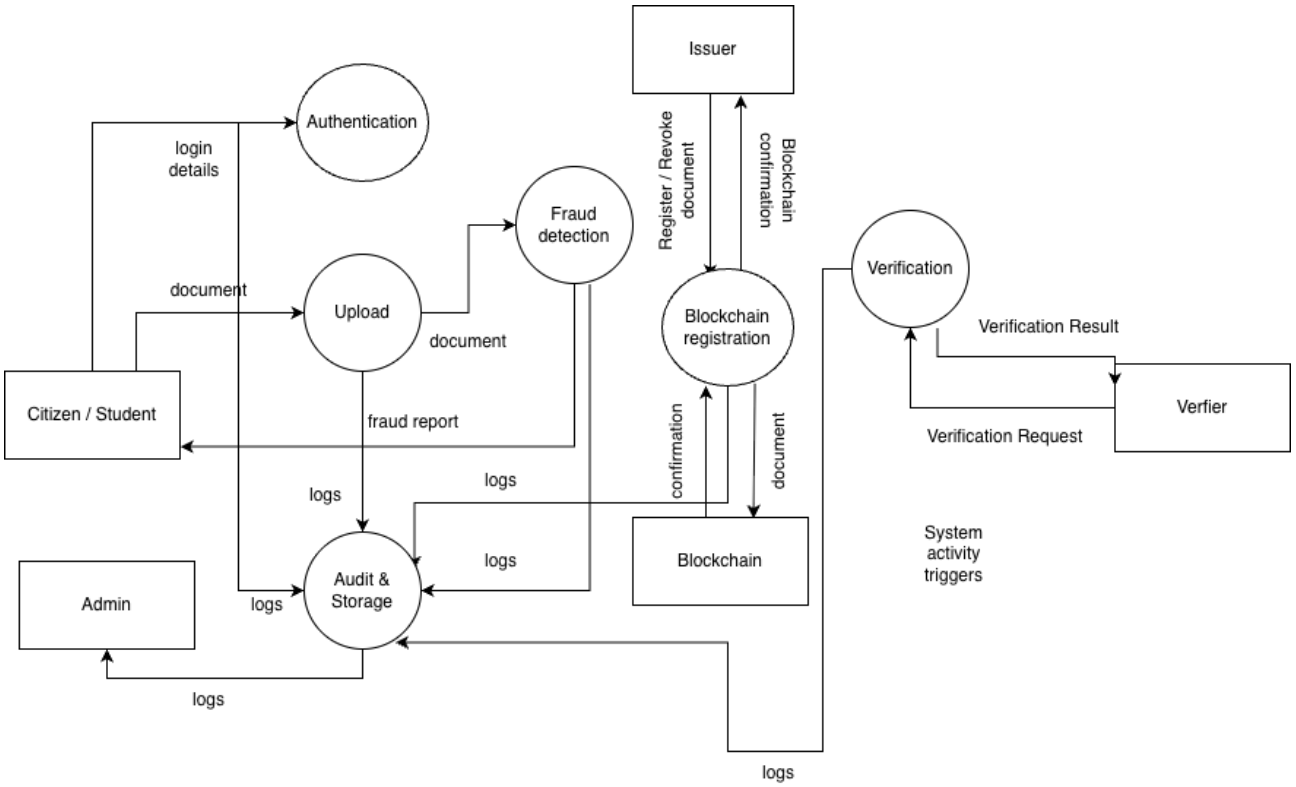
Task 6: Identify System Inputs and Outputs

Inputs	Outputs
User credentials [username, password, role]	Authentication response [login success/failed]
Uploaded documents	Authenticity score and fraud detection report
Document metadata	Verification result [Valid / Invalid / Tampered]
Verification requests	QR-based verification proof
Issuer actions	Blockchain transaction confirmation
User queries	Document Status
Extracted document data like text or metadata	Tampering alerts and anomaly detection results
System activity triggers	Audit logs of system activities
Role-based access requests	User notifications

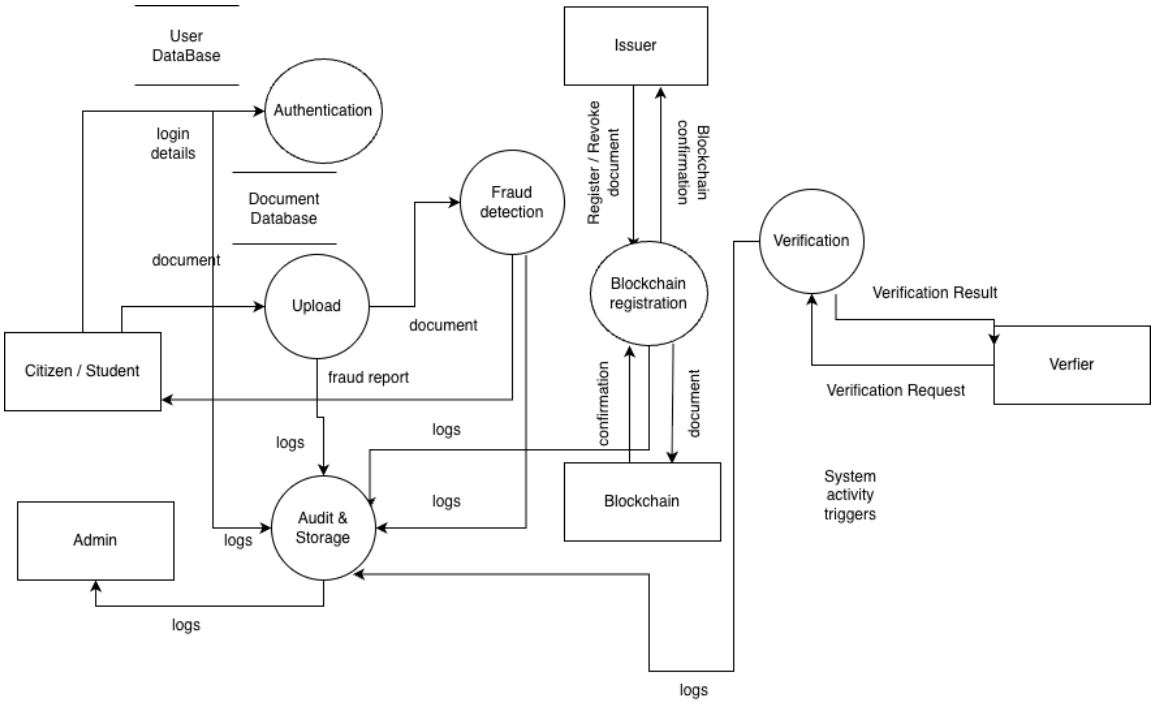
Task 7: Draw DFD Level 0 (Context Diagram)



Task 8: Draw DFD Level 1



Task 9: Draw DFD Level 2



PART D – CLASS DIAGRAM MODELING [CO04]

Task 10: Identify Classes

a. Identify all major classes required for the system.

b. Include:

1. Entity classes
2. Boundary classes
3. Control classes

Sl. No	Class Name	Type (Entity/Boundary/Control)	Description
1	User	Entity	Represents all system users
2	Document	Entity	Stores all uploaded documents
3	Metadata	Entity	Holds metadata of document
4	Hash	Entity	Stores the generated cryptographic hash
5	AuditLog	Entity	Tracks every action in the software
6	Blockchain	Entity	Stores immutable records of document hash
7	LoginInterface	Boundary	Handles user login interactions
8	DocumentInterface	Boundary	Handles interface to upload documents
9	LoginController	Control	Manages user authentication login
10	DocumentController	Control	Handles document upload and processing
11	VerificationController	Control	Handles document verification process
12	HashGenerator	Control	Generates hash for documents
13	AIAnalyzer	Control	Performs AI-based fraud detection

Task 11: Identify Attributes and Methods

For each class, list:

1. Attributes (with data types)
2. Methods (functions)

Class Name	Attributes	Methods
User	userID: String name: String role: String email: String password: String	login() logout() register()
Document	docID: String	upload()

	fileName: String uploadDate: Date ownerID: String	delete() getDetails()
Metadata	docType: String issuerName: String issueDate: Date	addMetadata() updateMetadata()
Hash	hashValue: String timestamp: Date	generateHash() validateHash()
AuditLog	recordID: String status: String verificationDate: Date	storeRecord() getHistory()
Blockchain	blockID: String previousHash: String currentHash: String	addBlock() verifyBlock()
LoginController	sessionID: String	authenticateUser() validateCredentials()
DocumentController	documentList: List	uploadDocument() processDocument()
VerificationController	verificationStatus: String	verifyDocument() compareHash()
HashGenerator	algorithm: String	generateHash()
AIAnalyzer	modelType: String	detectFraud() analyzeDocument()

Task 12: Identify Relationships

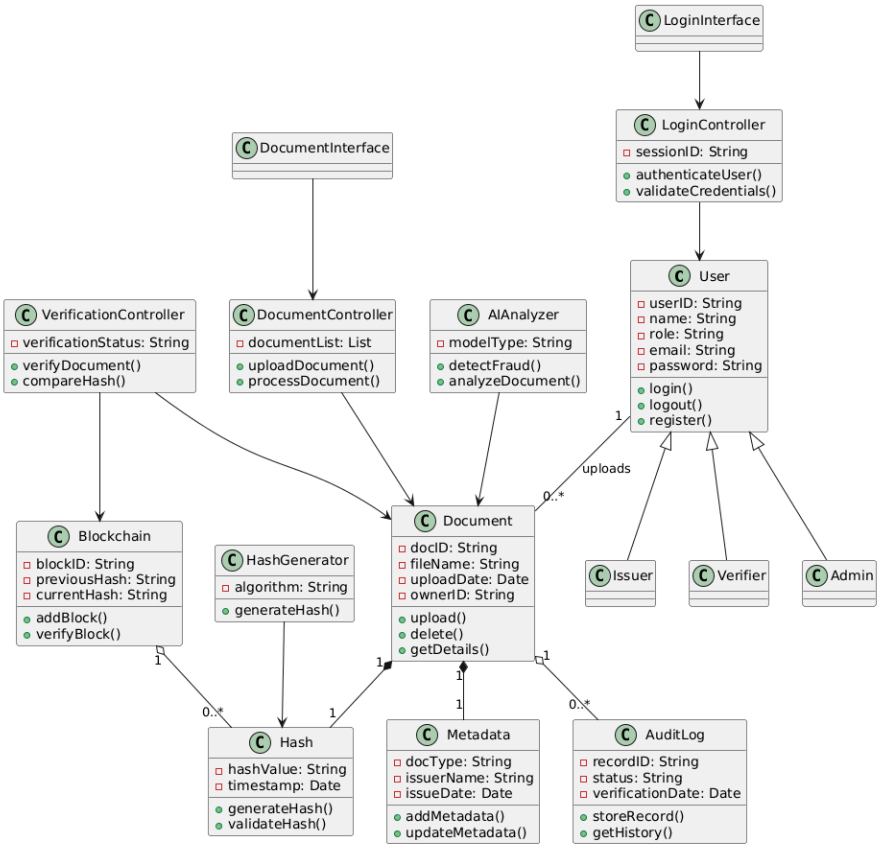
Identify relationships between classes:

1. Association
2. Aggregation
3. Composition
4. Inheritance

Class 1	Relationship	Class 2	Description
User	Association	Document	User uploads and owns documents
Document	Composition	Metadata	Metadata cannot exist without document
Document	Composition	Hash	Hash is generated for a

			document
Document	Aggregation	VerificationRecord	Document can have multiple verification records
VerificationController	Association	Document	Verifies documents
DocumentController	Association	Document	Manages document operations
AuthenticationController	Association	User	Handles login/authentication
HashGenerator	Association	Hash	Generates hash values
Blockchain	Aggregation	Hash	Stores hashes in blocks
VerificationController	Association	Blockchain	Checks stored hash
AIAnalyzer	Association	Document	Performs fraud analysis
User	Inheritance	Issuer / Verifier / Admin	Different user roles

Task 13: Draw Class Diagram



Task 14: Justification of Design

a. Why the classes were chosen
The classes were identified based on the main functionalities of the system such as user management, document handling, verification, and security. Entity classes like *User*, *Document*, and *Hash* represent core data objects. Boundary classes like *LoginInterface* and *DocumentInterface* manage user interaction. Control classes such as *DocumentController* and *VerificationControllerhandle* business logic. Additional classes like *Blockchain* and *AIAnalyzer* were included to support security and fraud detection features.

b. How relationships were identified
Relationships were determined based on how objects interact in the system workflow. Association was used where classes communicate, such as between *User* and *Document*. Composition was used where objects depend on a parent, like *Metadatabelonging* to *Document*. Aggregation was applied where objects can exist independently, such as *Blockchain* storing multiple hashes. Inheritance was used to represent different user roles like Admin, Issuer, and Verifier.

c. Any design assumptions made
It was assumed that each document has exactly one metadata and one hash value. Users can upload multiple documents but each document has a single owner. The blockchain stores hashes for verification but does not store full documents. AI-based fraud detection operates only during document verification. User roles such as Admin, Issuer, and Verifier share common properties inherited from the User class.